

Matrix Notation and Operations

Matrix Notation

An $r \times c$ *matrix* is a rectangular array of elements with r rows and c columns. An $r \times c$ matrix is said to be of *order* $r \times c$. A matrix is usually denoted by a capital letter printed in a boldface font (e.g., **A**, **B**, **X**). The elements of the matrix are represented by lower case letters with a double subscript (e.g., a_{jk} , b_{jk} , x_{jk}). For instance, in the matrix **X**, x_{13} is the element in the first row and the third column. A 4×3 matrix **X** is shown below.

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \\ x_{41} & x_{42} & x_{43} \end{bmatrix}$$

A matrix with a single row is called a *row vector* and a matrix with a single column is called a *column vector*. Vectors are usually represented by lower case letters printed in a boldface font (e.g., **a**, **b**, **x**). The elements of the vector are represented by lower case letters with a single subscript (e.g., a_j , b_j , x_j). A 3×1 column vector **y** and a 1×4 row vector **h** are shown below.

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\mathbf{h} = [h_1 \ h_2 \ h_3 \ h_4].$$

It is sometimes necessary to refer to a particular row or column of a matrix. These row or column vectors are represented by a subscripted lower case letter in a boldface font. For instance, the j^{th} row vector in the above 4×3 matrix **X** would be noted as **x_j** .

A *square matrix* has the same number of rows as columns. A square matrix where the *diagonal elements* (the elements where the two subscripts are equal) are nonzero and the *off-diagonal elements* are zero is called a *diagonal matrix*. A 3×3 diagonal matrix **D** is shown below.

$$\mathbf{D} = \begin{bmatrix} d_{11} & 0 & 0 \\ 0 & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix}$$

The *identity matrix* is a special type of diagonal matrix where all diagonal elements are equal to 1. The identity matrix is usually represented as $\mathbf{I}$ or $\mathbf{I}_n$ where n is the order of the identity matrix.

A square matrix where the jk^{th} element is equal to the kj^{th} element is called a *symmetric matrix*. A symmetric 3×3 matrix is shown below.

$$\mathbf{S} = \begin{bmatrix} 14 & 5 & 2 \\ 5 & 20 & 8 \\ 2 & 8 & 11 \end{bmatrix}$$

A *one vector* is a row or column vector in which every element is equal to 1 and is represented as the number one printed in a boldface font. A 1×3 one vector is shown below.

$$\mathbf{1} = [1 \quad 1 \quad 1]$$

Matrix Operations

The *transpose* of a matrix $\mathbf{X}$ is represented as $\mathbf{X}'$ (or $\mathbf{X}^T$). The transpose of a matrix is obtained by interchanging the rows and columns of the matrix. For instance, if

$$\mathbf{X} = \begin{bmatrix} 4 & 6 \\ 7 & 1 \\ 3 & 9 \end{bmatrix}$$

then

$$\mathbf{X}' = \begin{bmatrix} 4 & 7 & 3 \\ 6 & 1 & 9 \end{bmatrix}.$$

Note that the jk^{th} element in $\mathbf{X}$ is equal to the kj^{th} element in $\mathbf{X}'$. Most vectors in statistical formulas are assumed to be column vectors. Row vectors, when needed, are obtained by taking the transpose of a column vector.

If two matrices $\mathbf{A}$ and $\mathbf{B}$ are of the same order, the two matrices are then *conformable for addition or subtraction*, and $\mathbf{A} + \mathbf{B}$ is a matrix with element $a_{jk} + b_{jk}$ in the j^{th} row and the k^{th} column, as illustrated below for the sum of two 3×2 matrices.

$$\begin{aligned}\mathbf{A} + \mathbf{B} &= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \end{bmatrix}\end{aligned}$$

Likewise, $\mathbf{A} - \mathbf{B}$ is a matrix with element $a_{jk} - b_{jk}$ in the j^{th} row and the k^{th} column, as illustrated below for the difference of two 3×2 matrices.

$$\begin{aligned}\mathbf{A} - \mathbf{B} &= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} - \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} & a_{13} - b_{13} \\ a_{21} - b_{21} & a_{22} - b_{22} & a_{23} - b_{23} \end{bmatrix}\end{aligned}$$

To multiply a matrix by a *scalar* (i.e., a single number), simply multiply each element in the matrix by the scalar. Scalars are represented by italicized lower case letters in a non-boldface font. To illustrate, if $b = 2$ and

$$\mathbf{A} = \begin{bmatrix} 4 & 7 & 3 \\ 6 & 1 & 9 \end{bmatrix}$$

then

$$b\mathbf{A} = \begin{bmatrix} 8 & 14 & 6 \\ 12 & 2 & 18 \end{bmatrix}.$$

Some statistical formulas involve the subtraction of a scalar from an $n \times 1$ vector. The result is obtained by first multiplying the scalar by an $n \times 1$ one vector and then taking the difference of the two vectors. For instance, if $\mathbf{y}$ is a 3×1 vector and m is a scalar, then $\mathbf{y} - m$ is

$$\mathbf{y} - m\mathbf{1} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} - m \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} y_1 - m \\ y_2 - m \\ y_3 - m \end{bmatrix}$$

The *dot product* of a $n \times 1$ vector $\mathbf{a}$ with a $n \times 1$ vector $\mathbf{b}$ is

$$\mathbf{a}'\mathbf{b} = a_1b_1 + a_2b_2 + \dots + a_nb_n$$

Note that $\mathbf{a}'\mathbf{b} = \mathbf{b}'\mathbf{a}$. For instance, if $\mathbf{a}' = [4 \ 3 \ 2]$ and $\mathbf{b}' = [6 \ 1 \ 4]$, then $\mathbf{a}'\mathbf{b} = 4(6) + 3(1) + 2(4) = 35$.

Two $n \times 1$ vectors, $\mathbf{a}$ and $\mathbf{b}$, are said to be *orthogonal* if $\mathbf{a}'\mathbf{b} = 0$. For instance, if $\mathbf{a}' = [.5 \ .5 \ -1]$ and $\mathbf{b}' = [1 \ -1 \ 0]$, then $\mathbf{a}$ and $\mathbf{b}$ are orthogonal because $\mathbf{a}'\mathbf{b} = (.5)(1) + (.5)(-1) + (-1)(0) = 0$.

Two matrices $\mathbf{A}$ and $\mathbf{B}$ can be multiplied if they are *conformable for multiplication*. To compute the matrix product $\mathbf{AB}$, the number of columns of $\mathbf{A}$ must equal the number of rows of $\mathbf{B}$. In general, if $\mathbf{A}$ is $r \times n$ and $\mathbf{B}$ is $n \times c$, then the matrix product $\mathbf{AB}$ is an $r \times c$ matrix. The jk^{th} element in the $r \times c$ product matrix is equal to the dot product $\mathbf{a}_j\mathbf{b}_k$ where $\mathbf{a}_j$ is the j^{th} row vector of matrix $\mathbf{A}$ and $\mathbf{b}_k$ is the k^{th} column vector of matrix $\mathbf{B}$. For instance, the matrices $\mathbf{A}$ and $\mathbf{B}$ shown below are conformable for computing the product $\mathbf{AB}$ because $\mathbf{A}$ is 2×3 and $\mathbf{B}$ is 3×4 so that the product will be a 2×4 matrix.

$$\mathbf{A} = \begin{bmatrix} 4 & 7 & 3 \\ 6 & 1 & 9 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 5 & 4 & 3 & 1 \\ 4 & 2 & 3 & 2 \end{bmatrix}$$

Each of the $2 \times 4 = 8$ elements of the $\mathbf{AB}$ matrix is a dot product. For instance, the element in row 1 and column 1 of the product $\mathbf{AB}$ is

$$[4 \ 7 \ 3] \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix} = 4(1) + 7(5) + 3(4) = 51$$

and the element in row 2 and column 3 of $\mathbf{AB}$ is

$$[6 \ 1 \ 9] \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix} = 6(1) + 1(3) + 9(3) = 36.$$

After computing all 8 dot products, the following result is obtained.

$$\mathbf{AB} = \begin{bmatrix} 51 & 42 & 34 & 29 \\ 47 & 34 & 36 & 43 \end{bmatrix}$$

Unlike scalar multiplication where $ab = ba$, the matrix product $\mathbf{AB}$ does not in general equal $\mathbf{BA}$. Regarding the matrix product $\mathbf{AB}$, we can say that $\mathbf{B}$ is *pre-multiplied* by $\mathbf{A}$ or that $\mathbf{A}$ is *post-multiplied* by $\mathbf{B}$. The product of matrix $\mathbf{A}$ with itself is denoted as $\mathbf{A}^2$.

The transpose of a matrix product is equal to the product of the transposed matrices in reverse order. Specifically, $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$.

The product of three matrices $\mathbf{ABC}$ requires $\mathbf{A}$ and $\mathbf{B}$ to be conformable for multiplication and also requires $\mathbf{B}$ and $\mathbf{C}$ to be conformable for multiplication. The product $\mathbf{ABC}$ can be obtained by first computing $\mathbf{AB}$ and then post-multiplying the result by $\mathbf{C}$, or by first computing $\mathbf{BC}$ and then pre-multiplying the result by $\mathbf{A}$.

If $\mathbf{A}$ is a square matrix, then the *matrix inverse* of $\mathbf{A}$ is represented as $\mathbf{A}^{-1}$. If the inverse of $\mathbf{A}$ exists, then $\mathbf{AA}^{-1} = \mathbf{I}$. This result is a generalization of scalar arithmetic where $x(1/x) = 1$, assuming $x \neq 0$ so that the inverse of x exists. Computing a matrix inverse is tedious, and the amount of computational effort increases as the size of the matrix increases, but inverting a 2×2 matrix is not difficult. The inverse of a 2×2 matrix $\mathbf{A}$ is

$$\mathbf{A}^{-1} = (1/d) \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

where $d = a_{11}a_{22} - a_{12}a_{21}$ is called the *determinant* of $\mathbf{A}$. The matrix inverse does not exist unless the determinant is nonzero.

Inverting a diagonal matrix $\mathbf{D}$ of any order is simple. The inverse of $\mathbf{D}$ is equal to a diagonal matrix where the j^{th} diagonal element is equal to $1/d_j$.

The *trace* of a square $n \times n$ matrix $\mathbf{A}$, denoted as $tr(\mathbf{A})$, is defined as the sum of its n diagonal elements.

$$tr(\mathbf{A}) = a_{11} + a_{22} + \dots + a_{nn}$$

For instance if

$$\mathbf{V} = \begin{bmatrix} 14 & 5 & 2 \\ 9 & 20 & 8 \\ 7 & 8 & 11 \end{bmatrix}$$

then $\text{tr}(\mathbf{V}) = 14 + 20 + 11 = 45$.

The *Kronecker product* of two matrices, an $m \times n$ matrix $\mathbf{A}$ and a $p \times q$ matrix $\mathbf{B}$, is defined to be the $mp \times nq$ matrix

$$\mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2n}\mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & a_{m2}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{bmatrix}$$

which is obtained by replacing each element a_{jk} with the $p \times q$ matrix $a_{jk}\mathbf{B}$. For example, if

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & -1 \end{bmatrix} \text{ and } \mathbf{b} = [1 \ 2 \ 3]$$

then

$$\mathbf{A} \otimes \mathbf{b} = \begin{bmatrix} 1 & 2 & 3 & 2 & 4 & 6 \\ 3 & 6 & 9 & -1 & -2 & -3 \end{bmatrix}.$$

The Kronecker product of an identity matrix and another matrix has the following simple form

$$\mathbf{I} \otimes \mathbf{B} = \begin{bmatrix} \mathbf{B} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{B} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{B} \end{bmatrix}$$

where $\mathbf{0}$ is a matrix of zeros that has the same order as $\mathbf{B}$.

The transpose of a Kronecker product of two matrices is

$$(\mathbf{A} \otimes \mathbf{B})' = \mathbf{A}' \otimes \mathbf{B}'$$

and the inverse of a Kronecker product of two matrices is

$$(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}.$$

The product of $\mathbf{A} \otimes \mathbf{B}$ and $\mathbf{C} \otimes \mathbf{D}$ is equal to

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$$

assuming $\mathbf{A}$ and $\mathbf{C}$ are conformable for multiplication and $\mathbf{B}$ and $\mathbf{D}$ are conformable for multiplication.

In some statistical formulas, it is convenient to rearrange the elements of an $r \times c$ matrix $\mathbf{A}$ into an $rc \times 1$ column vector $\mathbf{a}$. This is done by stacking the c column vectors (which are each $r \times 1$) of $\mathbf{A}$ ($\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_c$) one under the other as shown below

$$\mathbf{a} = \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_c \end{bmatrix}$$

The conversion of a matrix $\mathbf{A}$ into a vector is denoted as $vec(\mathbf{A})$. To illustrate, consider the following 4×2 matrix $\mathbf{Y}$ and $vec(\mathbf{Y})$. The $mat(\mathbf{y})$ function returns the vector into its original matrix form.

$$\mathbf{Y} = \begin{bmatrix} 14 & 2 \\ 9 & 8 \\ 7 & 11 \\ 16 & 34 \end{bmatrix} \quad vec(\mathbf{Y}) = \begin{bmatrix} 14 \\ 9 \\ 7 \\ 16 \\ 2 \\ 8 \\ 11 \\ 34 \end{bmatrix} = \mathbf{y}$$

The vectorization of a matrix triple product can be expressed as

$$vec(\mathbf{ABC}) = (\mathbf{C}' \otimes \mathbf{A})vec(\mathbf{B})$$

and the vectorization of a matrix product $\mathbf{AB}$ follows from this result by setting $\mathbf{C} = \mathbf{I}$

$$\text{vec}(\mathbf{AB}) = (\mathbf{I}_p \otimes \mathbf{A})\text{vec}(\mathbf{B})$$

where p is the number of columns of $\mathbf{B}$.

For two matrices $\mathbf{A}$ and $\mathbf{B}$ of the same order, *matrix equality* $\mathbf{A} = \mathbf{B}$ indicates that $a_{jk} = b_{jk}$ for every value of j and k . *Matrix inequality* $\mathbf{A} \neq \mathbf{B}$ indicates that there is at least one element in $\mathbf{A}$ that is not equal to its corresponding element in $\mathbf{B}$.

Covariance Matrices

A *covariance matrix* is a symmetric matrix with variances in the diagonal elements and covariances in the off-diagonal elements. If r response variables $\mathbf{y}' = [y_1 \ y_2 \ \dots \ y_r]$ are measured for each person in a random sample of n people, the estimated variance for the j^{th} response variable is

$$\hat{\sigma}_j^2 = \sum_{i=1}^n (y_{ij} - \hat{\mu}_j)^2 / (n - 1)$$

and the estimated covariance between the j^{th} and k^{th} response variables is

$$\hat{\sigma}_{jk} = \sum_{i=1}^n (y_{ij} - \hat{\mu}_j)(y_{ik} - \hat{\mu}_k) / (n - 1).$$

The estimated covariance between the j^{th} and k^{th} measurements is also equal to $\hat{\rho}_{jk}\hat{\sigma}_j\hat{\sigma}_k$ where $\hat{\rho}_{jk}$ is the estimated Pearson correlation between the two response variables. Note that $\hat{\sigma}_{jk} = \hat{\sigma}_{kj}$.

The r variances and the $r(r - 1)/2$ covariances of $\mathbf{y}' = [y_1 \ y_2 \ \dots \ y_r]$ can be summarized in an $r \times r$ covariance matrix denoted as $\text{cov}(\mathbf{y})$. For instance, with $r = 3$ the covariance matrix is

$$\text{cov}(\mathbf{y}) = \begin{bmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} & \hat{\sigma}_{13} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 & \hat{\sigma}_{23} \\ \hat{\sigma}_{13} & \hat{\sigma}_{23} & \hat{\sigma}_3^2 \end{bmatrix}.$$

If there are n sets of response variables where the i^{th} set has a covariance matrix $\mathbf{S}_i$, and response variables from different sets are assumed to be uncorrelated, the covariance matrix for all response variables has the following form

$$\begin{bmatrix} \mathbf{S}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{S}_n \end{bmatrix}$$

where each $\mathbf{0}$ represents a matrix of zeros. If all of the n covariance matrices are equal to $\mathbf{S}$, the above covariance matrix can then be expressed as $\mathbf{I} \otimes \mathbf{S}$.

Variance of a Linear Function of Variables

A linear function of r variables can be expressed as $\sum_{j=1}^r h_j y_j$ where the h_j coefficients are known numbers. This linear function can be expressed in matrix notation as $\mathbf{h}'\mathbf{y}$ where $\mathbf{h}' = [h_1 \ h_2 \ \dots \ h_r]$ and $\mathbf{y}' = [y_1 \ y_2 \ \dots \ y_r]$. The variance of $\mathbf{h}'\mathbf{y}$ can be expressed in matrix notation as $\mathbf{h}'\text{cov}(\mathbf{y})\mathbf{h}$. For instance, the variance of $y_1 + y_2 + y_3$ is

$$\begin{aligned} & [1 \ 1 \ 1] \begin{bmatrix} \hat{\sigma}_1^2 & \hat{\sigma}_{12} & \hat{\sigma}_{13} \\ \hat{\sigma}_{12} & \hat{\sigma}_2^2 & \hat{\sigma}_{23} \\ \hat{\sigma}_{13} & \hat{\sigma}_{23} & \hat{\sigma}_3^2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\ &= [\hat{\sigma}_1^2 + \hat{\sigma}_{12} + \hat{\sigma}_{13} \quad \hat{\sigma}_{12} + \hat{\sigma}_2^2 + \hat{\sigma}_{23} \quad \hat{\sigma}_{13} + \hat{\sigma}_{23} + \hat{\sigma}_3^2] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\ &= \hat{\sigma}_1^2 + \hat{\sigma}_2^2 + \hat{\sigma}_3^2 + 2\hat{\sigma}_{12} + 2\hat{\sigma}_{13} + 2\hat{\sigma}_{23}. \end{aligned}$$

The variance of a linear function of parameter estimates (i.e., $\mathbf{c}'\hat{\boldsymbol{\beta}}$) is similarly defined as $\mathbf{c}'\text{cov}(\hat{\boldsymbol{\beta}})\mathbf{c}$. For instance, the variance of $\hat{\beta}_1 - \hat{\beta}_2$ is

$$\begin{aligned} & [1 \ -1] \begin{bmatrix} \hat{\sigma}_{\hat{\beta}_1}^2 & \hat{\sigma}_{\hat{\beta}_1\hat{\beta}_2} \\ \hat{\sigma}_{\hat{\beta}_1\hat{\beta}_2} & \hat{\sigma}_{\hat{\beta}_2}^2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = [\hat{\sigma}_{\hat{\beta}_1}^2 - \hat{\sigma}_{\hat{\beta}_1\hat{\beta}_2} \quad \hat{\sigma}_{\hat{\beta}_1\hat{\beta}_2} - \hat{\sigma}_{\hat{\beta}_2}^2] \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \hat{\sigma}_{\hat{\beta}_1}^2 - \hat{\sigma}_{\hat{\beta}_1\hat{\beta}_2} - \hat{\sigma}_{\hat{\beta}_1\hat{\beta}_2} + \hat{\sigma}_{\hat{\beta}_2}^2 \\ &= \hat{\sigma}_{\hat{\beta}_1}^2 + \hat{\sigma}_{\hat{\beta}_2}^2 - 2\hat{\sigma}_{\hat{\beta}_1\hat{\beta}_2}. \end{aligned}$$

General Linear Model in Matrix Form

A general linear model (GLM) model for a random sample of n participants can be expressed in matrix notation as follows

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e} \quad (1.1)$$

where $\mathbf{y}$ is a $n \times 1$ vector of response variable scores, $\mathbf{e}$ is an $n \times 1$ vector of prediction errors with $\text{cov}(\mathbf{e}) = \sigma_e^2 \mathbf{I}_n$, $\boldsymbol{\beta}$ is a $(q + 1) \times 1$ vector of unknown population parameters, and $\mathbf{X}$ is an $n \times (q + 1)$ *design matrix*. The first column of $\mathbf{X}$ is often an $n \times 1$ vector of ones to code the y -intercept and other q columns contain the values of q predictor variables.

To illustrate the structure of Equation 1.1, consider a model with $q = 1$ predictor variable and a random sample of $n = 7$. The elements of $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}$ for this example are shown below.

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \end{bmatrix} = \begin{bmatrix} 1 & x_{11} \\ 1 & x_{12} \\ 1 & x_{13} \\ 1 & x_{14} \\ 1 & x_{15} \\ 1 & x_{16} \\ 1 & x_{17} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \end{bmatrix}$$

Using matrix notation, the *ordinary least squares* (OLS) estimate of $\boldsymbol{\beta}$ is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} \quad (1.2)$$

and the estimated residuals are

$$\hat{\mathbf{e}} = \mathbf{y} - \hat{\mathbf{y}} \quad (1.3)$$

where $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is a vector of predicted y scores. An estimate of the residual variance (σ_e^2) is

$$MS_E = \hat{\mathbf{e}}'\hat{\mathbf{e}}/(n - q - 1). \quad (1.4)$$

The j^{th} element of $\hat{\boldsymbol{\beta}}$ is $\hat{\beta}_j$. The standard error of $\hat{\beta}_j$ is equal to the square root of the j^{th} diagonal element of the following covariance matrix of $\hat{\boldsymbol{\beta}}$

$$\text{cov}(\hat{\boldsymbol{\beta}}) = MS_E(\mathbf{X}'\mathbf{X})^{-1}. \quad (1.5)$$

A linear function of the elements of $\boldsymbol{\beta}$ will be of interest in some applications. A linear function of $\boldsymbol{\beta}$ can be expressed as $\mathbf{c}'\boldsymbol{\beta}$ where $\mathbf{c}$ is a $(q + 1) \times 1$ vector of specified numbers.

For instance, in a GLM with $q = 2$ predictor variables the linear function $\beta_1 + 5\beta_2$ can be expressed as $\mathbf{c}'\boldsymbol{\beta}$ where $\mathbf{c}' = [0 \ 1 \ 5]$. The estimated standard error of $\mathbf{c}'\hat{\boldsymbol{\beta}}$ is

$$SE_{\mathbf{c}'\hat{\boldsymbol{\beta}}} = \sqrt{\mathbf{c}' \text{cov}(\hat{\boldsymbol{\beta}}) \mathbf{c}} \quad (1.6)$$

and a $100(1 - \alpha)\%$ confidence interval for $\mathbf{c}'\boldsymbol{\beta}$ is

$$\mathbf{c}'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2; df_E} SE_{\mathbf{c}'\hat{\boldsymbol{\beta}}} \quad (1.7)$$

where $df_E = n - q - 1$. To obtain simultaneous Bonferroni confidence intervals for v different linear functions of $\boldsymbol{\beta}$, replace α with $\alpha^* = \alpha/v$ in Equation 1.7.

A special case of Equation 1.7 is the following confidence interval for the j^{th} element in $\boldsymbol{\beta}$

$$\hat{\beta}_j \pm t_{\alpha/2; df_E} SE_{\hat{\beta}_j} \quad (1.8)$$

where

$$SE_{\hat{\beta}_j} = \sqrt{\frac{MS_E}{(1 - \hat{\rho}_{x_j, \mathbf{x}}^2) \hat{\sigma}_{x_j}^2 (n - 1)}} \quad (1.9)$$

and $\hat{\rho}_{x_j, \mathbf{x}}^2$ is the estimated squared multiple correlation between predictor variable j and all other predictor variables.

Another special case of Equation 1.7 is the following confidence interval for the difference between any two elements in $\boldsymbol{\beta}$, such as β_1 and β_2

$$\hat{\beta}_1 - \hat{\beta}_2 \pm t_{\alpha/2; df_E} \sqrt{SE_{\hat{\beta}_1}^2 + SE_{\hat{\beta}_2}^2 + \text{cov}(\hat{\beta}_1, \hat{\beta}_2)} \quad (1.10)$$

where $\text{cov}(\hat{\beta}_1, \hat{\beta}_2)$ is obtained from Equation 1.5.

In a random-x GLM (where all q predictor variables are random), the squared multiple correlation (denoted as $\rho_{y, \mathbf{x}}^2$) describes the proportion of variance in the y scores that is predictable from all q predictor variables. An estimate of $\rho_{y, \mathbf{x}}^2$ (reported as “R-squared” in most statistical packages) is

$$\hat{\rho}_{y, \mathbf{x}}^2 = 1 - SS_E / SS_T. \quad (1.11)$$

where $SS_E = \hat{\mathbf{e}}' \hat{\mathbf{e}}$ and $SS_T = (n - 1) \hat{\sigma}_y^2$. The fact that the multiple correlation ($\rho_{y.x}$) is equal to the Pearson correlation between y and the predicted y scores is helpful in interpreting the multiple correlation and squared multiple correlation. A confidence interval for $\rho_{y.x}^2$ does not have a simple formula but it can be computed using SAS or R.

In a fixed- $\mathbf{x}$ GLM (where all the predictor variables are fixed), the following estimate of the coefficient of multiple determination

$$\hat{\eta}^2 = 1 - SS_E/SS_T \quad (1.12)$$

is an estimate of η^2 which is equal to $\hat{\rho}_{y.x}^2$. Like $\hat{\rho}_{y.x}^2$, $\hat{\eta}^2$ has a positive bias and the bias can be substantial when $n - q$ is small. A less biased estimate of η^2 (confusingly called omega-squared) is available in SAS. Although $\eta^2 = \rho_{y.x}^2$ and $\hat{\eta}^2 = \hat{\rho}_{y.x}^2$, different symbols are used in the random- $\mathbf{x}$ and fixed- $\mathbf{x}$ models because $\hat{\eta}^2$ and $\hat{\rho}_{y.x}^2$ have different sampling distributions, and a confidence interval for η^2 in the fixed- $\mathbf{x}$ model will be different than a confidence interval for $\rho_{y.x}^2$ in the random- $\mathbf{x}$ model. The confidence interval for η^2 is complicated but it can be obtained in SAS or R.

Multivariate General Linear Model

Some studies will involve $r \geq 2$ response variables. A GLM for a random sample of n participants, with the same set of q predictor variables for each response variable, can be specified for each of the r response variables

$$\mathbf{y}_1 = \mathbf{X}\boldsymbol{\beta}_1 + \mathbf{e}_1 \quad \mathbf{y}_2 = \mathbf{X}\boldsymbol{\beta}_2 + \mathbf{e}_2 \quad \dots \quad \mathbf{y}_r = \mathbf{X}\boldsymbol{\beta}_r + \mathbf{e}_r$$

and these r models can be combined into the following *multivariate general linear model* (MGLM)

$$\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E}$$

where $\mathbf{Y} = [\mathbf{y}_1 \ \mathbf{y}_2 \ \dots \ \mathbf{y}_r]$ is an $n \times r$ matrix of response variables, $\mathbf{X}$ is an $n \times (q + 1)$ design matrix, $\mathbf{B} = [\boldsymbol{\beta}_1 \ \boldsymbol{\beta}_2 \ \dots \ \boldsymbol{\beta}_r]$ is a $(q + 1) \times r$ matrix of unknown parameters, and $\mathbf{E} = [\mathbf{e}_1 \ \mathbf{e}_2 \ \dots \ \mathbf{e}_r]$ is an $n \times r$ matrix of prediction errors. Note that the design matrix in the MGLM is the same as the design matrix in each GLM. The r columns of $\mathbf{E}$ are assumed to be correlated and the $r \times r$ covariance matrix of the r columns of $\mathbf{E}$ will be denoted as $\mathbf{S}$.

The independence assumption of the GLM is also required in the MGLM and this assumption implies that the rows of $\mathbf{E}$ are uncorrelated.

The OLS estimate of $\mathbf{B}$ in the MGLM is

$$\hat{\mathbf{B}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}, \quad (1.17)$$

and the matrix of predicted scores is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\mathbf{B}}. \quad (1.18)$$

The matrix of estimated prediction errors (residuals) is

$$\hat{\mathbf{E}} = \mathbf{Y} - \hat{\mathbf{Y}}, \quad (1.19)$$

and an estimate of the covariance matrix $\mathbf{S}$ is

$$\hat{\mathbf{S}} = \hat{\mathbf{E}}'\hat{\mathbf{E}}/(n - q - 1). \quad (1.20)$$

Let $\hat{\boldsymbol{\beta}}^* = \text{vec}(\hat{\mathbf{B}})$. The covariance matrix of $\hat{\boldsymbol{\beta}}^*$ is

$$\begin{aligned} \text{cov}(\hat{\boldsymbol{\beta}}^*) &= \left[(\mathbf{I}_r \otimes \mathbf{X})' (\hat{\mathbf{S}} \otimes \mathbf{I}_n)^{-1} (\mathbf{I}_r \otimes \mathbf{X}) \right]^{-1} \\ &= \hat{\mathbf{S}} \otimes (\mathbf{X}'\mathbf{X})^{-1}. \end{aligned} \quad (1.21)$$

A *multivariate linear contrast* of the $\mathbf{B}$ parameter matrix in the MGLM can be expressed as $\mathbf{c}'\mathbf{B}\mathbf{h}$ where $\mathbf{c}$ is a $(q + 1) \times 1$ vector of researcher-specified coefficients and $\mathbf{h}$ is an $r \times 1$ vector of researcher-specified coefficients. To illustrate the specification of multivariate linear contrasts, consider the parameter matrix of the MGLM given above with $q = 3$ predictor variables and $r = 2$ response variables. In this model, $\mathbf{B}$ has four rows and two columns. Each column of $\mathbf{B}$ contains the y -intercept and the three slope coefficients for a single response variable. The 4×2 parameter matrix for this model is given below.

$$\mathbf{B} = \begin{bmatrix} \beta_{01} & \beta_{02} \\ \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \\ \beta_{31} & \beta_{32} \end{bmatrix}$$

Some examples will illustrate the specification of $\mathbf{c}$ and $\mathbf{h}$. To estimate $\beta_{11} - \beta_{21}$, set $\mathbf{c}' = [0 \ 1 \ -1 \ 0]$ and $\mathbf{h}' = [1 \ 0]$. To estimate $\beta_{11} - \beta_{12}$, set $\mathbf{c}' = [0 \ 1 \ 0 \ 0]$ and $\mathbf{h}' = [1 \ -1]$. To estimate $(\beta_{11} - \beta_{21}) - (\beta_{21} - \beta_{22})$, set $\mathbf{c}' = [0 \ 1 \ -1 \ 0]$ and $\mathbf{h}' = [1 \ -1]$.

A $100(1 - \alpha)\%$ confidence interval for $\mathbf{c}'\mathbf{B}\mathbf{h}$ is

$$\mathbf{c}'\hat{\mathbf{B}}\mathbf{h} \pm t_{\alpha/2; df} \sqrt{(\mathbf{h}'\hat{\mathbf{S}}\mathbf{h})\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}} \quad (1.22)$$

where $df = n - q - 1$ and $\sqrt{(\mathbf{h}'\hat{\mathbf{S}}\mathbf{h})\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}$ is the estimated standard error of $\mathbf{c}'\hat{\mathbf{B}}\mathbf{h}$. To obtain simultaneous Bonferroni confidence intervals for v different multivariate linear contrasts of $\mathbf{B}$, replace α with $\alpha^* = \alpha/v$ in Equation 1.22. A confidence interval for $\mathbf{c}'\mathbf{B}\mathbf{h}$ can be used to test $H_0: \mathbf{c}'\mathbf{B}\mathbf{h} = 0$ against $H_1: \mathbf{c}'\mathbf{B}\mathbf{h} > 0$ and $H_2: \mathbf{c}'\mathbf{B}\mathbf{h} < 0$ using the following three-decision rule:

- If the upper limit of the confidence interval for $\mathbf{c}'\mathbf{B}\mathbf{h}$ is less than 0, then H_0 is rejected and $H_2: \mathbf{c}'\mathbf{B}\mathbf{h} < 0$ is accepted.
- If the lower limit of the confidence interval for $\mathbf{c}'\mathbf{B}\mathbf{h}$ is greater than 0, then H_0 is rejected and $H_1: \mathbf{c}'\mathbf{B}\mathbf{h} > 0$ is accepted.
- If the confidence interval for $\mathbf{c}'\mathbf{B}\mathbf{h}$ includes 0, then H_0 cannot be rejected and the results are inconclusive.

Some computer programs will compute the test statistic $\mathbf{c}'\hat{\mathbf{B}}\mathbf{h}/\sqrt{(\mathbf{h}'\hat{\mathbf{S}}\mathbf{h})\mathbf{c}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{c}}$ and its associated p -value. The p -value is used to decide if H_0 can be rejected.

Although Equations 1.17 - 1.22 are elegant in their generality, all of these results can be obtained from GLM results. Specifically, the j^{th} column of $\hat{\mathbf{B}}$ in Equation 1.17 can be computed using $\hat{\boldsymbol{\beta}}_j = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}_j$ (Equation 1.2), and for each column of $\hat{\mathbf{B}}$, the standard errors can be computed from $MSE_j(\mathbf{X}'\mathbf{X})^{-1}$ (Equation 1.5) where MSE_j is the j^{th} diagonal element of $\hat{\mathbf{S}}$. Thus, the least squares estimates of $\mathbf{B}$ and their standard errors can be obtained by simply analyzing each of the r response variables separately using a GLM. To obtain the results for Equation 1.22 using GLM results, it can be shown that the confidence interval for $\mathbf{c}'\boldsymbol{\beta}$ (Equation 1.7) is identical to Equation 1.22 if the response variable in the GLM is defined as $y = h_1y_1 + h_2y_2 + \dots + h_ry_r$.

Constrained MGLM

The MGLM can be expressed in the following *vector form*

$$\mathbf{y}^* = \mathbf{X}^* \boldsymbol{\beta}^* + \mathbf{e}^*$$

where $\mathbf{y}^* = \text{vec}(\mathbf{Y})$, $\mathbf{X}^* = (\mathbf{I}_r \otimes \mathbf{X})$, $\boldsymbol{\beta}^* = \text{vec}(\mathbf{B})$, $\mathbf{e}^* = \text{vec}(\mathbf{E})$, and $\text{cov}(\mathbf{e}^*) = \mathbf{S} \otimes \mathbf{I}_n$. Note that the order of $\boldsymbol{\beta}^*$ is $r(q+1) \times 1$ and $\mathbf{X}^*$ has the following block-diagonal form

$$\mathbf{X}^* = \begin{bmatrix} \mathbf{X} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{X} & \cdots & \mathbf{0} \\ \vdots & \vdots & & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{X} \end{bmatrix}.$$

This design matrix for the vector form of the MGLM shows that all q predictor variables are related to all r response variables. Now suppose the predictor variables are not the same for each response variable. A MGLM with one or more slope coefficients constrained to equal 0 is referred to in the statistics and economics literature as a *seemingly unrelated regression (SUR) model*.

With response variable y_j having its own set of q_j predictor variables, the SUR model can be expressed as $\mathbf{y}^* = \mathbf{X}^* \boldsymbol{\beta}^* + \mathbf{e}^*$ where $\boldsymbol{\beta}^*$ is a $(q_1 + q_2 + \dots + q_r + r) \times 1$ vector and $\mathbf{X}^*$ has the following block structure form

$$\mathbf{X}^* = \begin{bmatrix} \mathbf{X}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{X}_r \end{bmatrix}.$$

The prediction errors in the SUR model have the covariance structure $\text{cov}(\mathbf{e}^*) = \mathbf{S} \otimes \mathbf{I}_n$ and the following *generalized least squares* (GLS) estimate of $\boldsymbol{\beta}^*$ is typically used

$$\hat{\boldsymbol{\beta}}_{GLS}^* = [\mathbf{X}^{*'} (\hat{\mathbf{S}}^{-1} \otimes \mathbf{I}_n) \mathbf{X}^*]^{-1} \mathbf{X}^{*'} (\hat{\mathbf{S}}^{-1} \otimes \mathbf{I}_n) \mathbf{y}^* \quad (1.23)$$

where $\hat{\mathbf{S}}$ is an estimate of $\mathbf{S}$. The traditional method of estimating $\mathbf{S}$ is to first compute a OLS estimate of $\boldsymbol{\beta}^*$ ($\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^{*'} \mathbf{X}^*)^{-1} \mathbf{X}^{*'} \mathbf{y}^*$), compute the vector of estimated prediction errors ($\hat{\mathbf{e}} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}^*$), convert the vector $\hat{\mathbf{e}}$ into its matrix form ($\hat{\mathbf{E}} = \text{mat}(\hat{\mathbf{e}})$), and then estimate $\mathbf{S}$ as

$$\hat{\mathbf{S}} = \mathbf{D} \hat{\mathbf{E}}' \hat{\mathbf{E}} \mathbf{D} \quad (1.24)$$

where $\mathbf{D}$ is a diagonal matrix with $\sqrt{1/(n - q_j - 1)}$ in the j^{th} diagonal position and q_j is the number of predictors of response variable j . Equations 1.20 and 1.24 are unstructured covariance matrices with r estimated variances and $r(r - 1)/2$ estimated covariances.

The estimated covariance matrix of $\hat{\boldsymbol{\beta}}_{GLS}^*$ is

$$\text{cov}(\hat{\boldsymbol{\beta}}_{GLS}^*) = [\mathbf{X}'(\hat{\mathbf{S}}^{-1} \otimes \mathbf{I}_n)\mathbf{X}]^{-1} \quad (1.25)$$

and the square roots of the diagonal elements of $\text{cov}(\hat{\boldsymbol{\beta}}_{GLS}^*)$ are the standard errors of the GLS estimates. An approximate $100(1 - \alpha)\%$ confidence interval for $\mathbf{a}'\boldsymbol{\beta}^*$ is

$$\mathbf{a}'\hat{\boldsymbol{\beta}}_{GLS}^* \pm z_{\alpha/2} \sqrt{\mathbf{a}'\text{cov}(\hat{\boldsymbol{\beta}}_{GLS}^*)\mathbf{a}} \quad (1.26)$$

and an approximate confidence interval for β_j^* (the j^{th} element in $\boldsymbol{\beta}^*$) is obtained by specifying the j^{th} element of $\mathbf{a}$ to equal 1 and all other values equal to 0. Equation 1.26 is approximate because it uses a GLS estimate of $\boldsymbol{\beta}^*$ that is biased in small samples and also because the sampling distribution of $\mathbf{a}'\hat{\boldsymbol{\beta}}_{GLS}^*$ is not accurately approximated by a normal distribution in small samples.